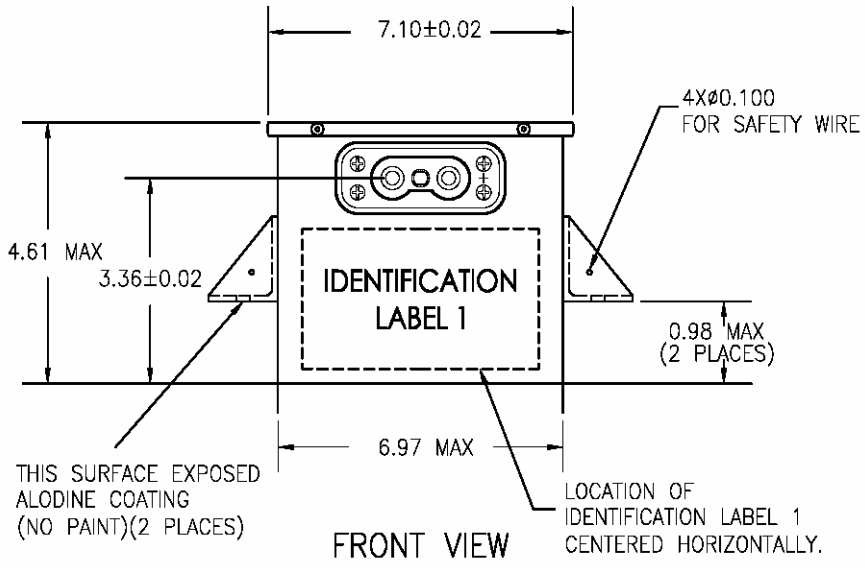
INCH-POUND
MIL-PRF-8565/16
26 December 2006

PERFORMANCE SPECIFICATION SHEET

BATTERY, STORAGE, AIRCRAFT, MEDIUM RATE,
MAINTENANCE-FREE, 24-VOLT, 5-AMPERE-HOUR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-8565.



Notes:
1. All dimension are in inches.
2. Abbreviation used in this figure are maximum (MAX), minimum (MIN) and diameter (Ø)

FIGURE 1. Dimensions and configuration for PIN M8565/16-1.

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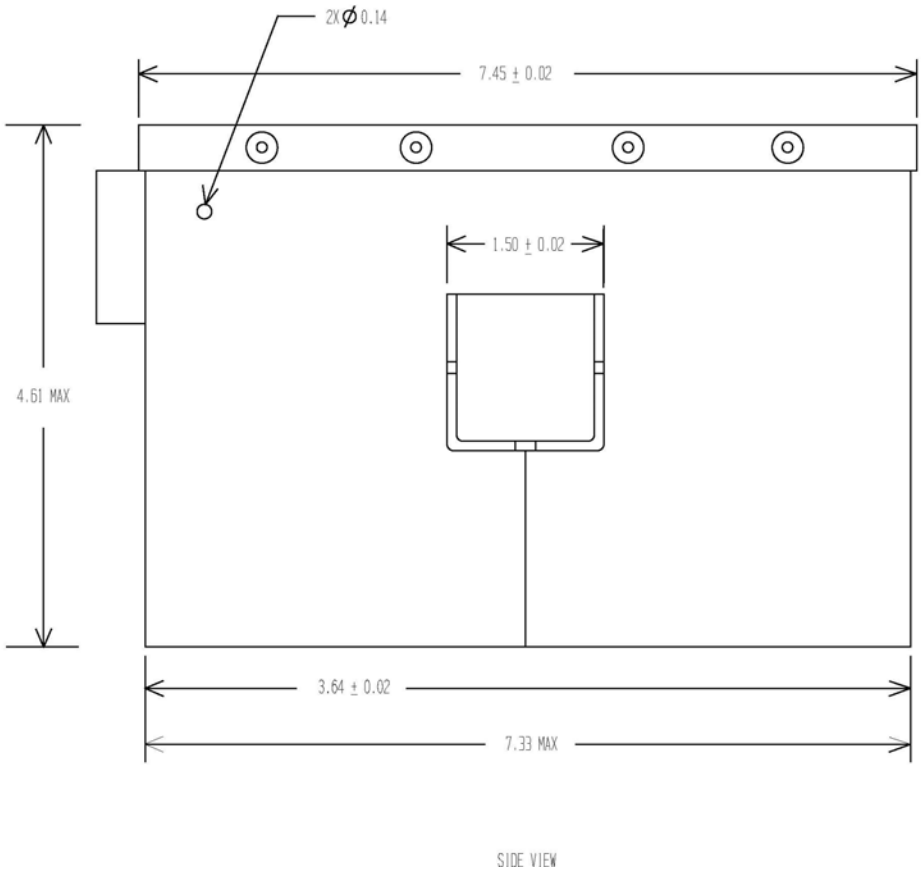
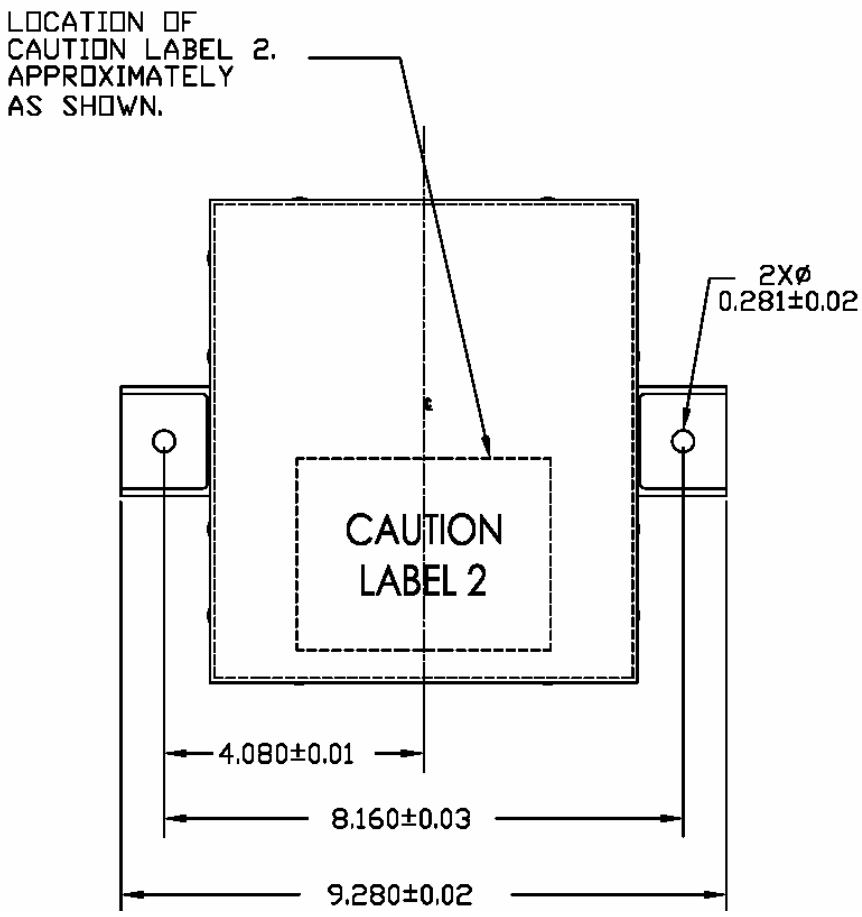


FIGURE 1. Dimensions and configuration for PIN M8565/16-1 - Continued.

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Top View

FIGURE 1. Dimensions and configuration for PIN M8565/16-1 - Continued.

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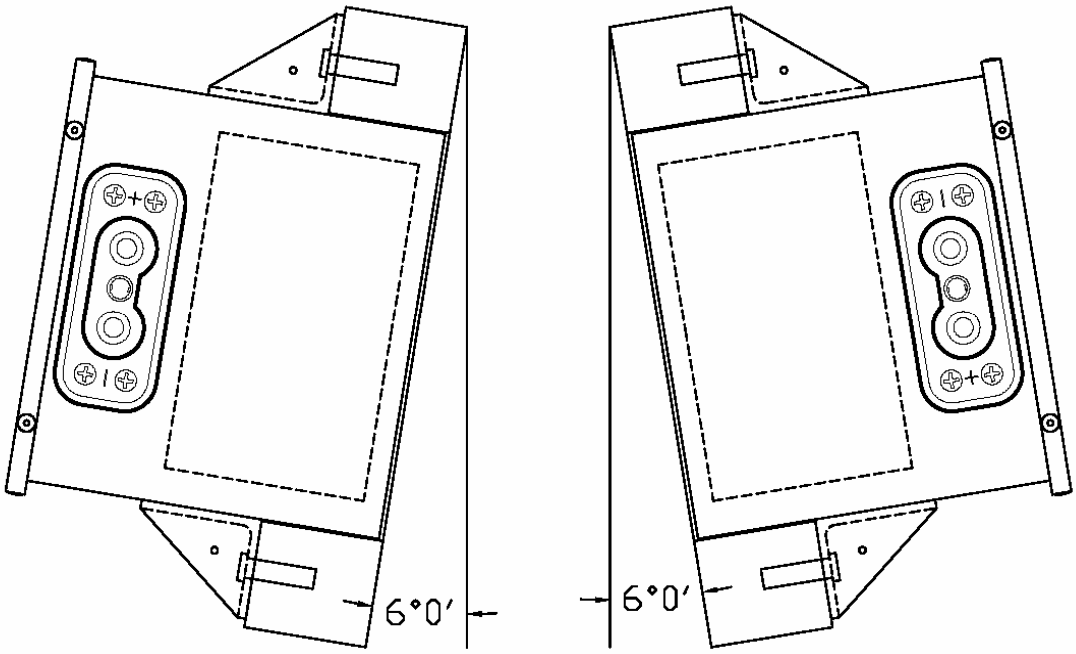


FIGURE 2. Typical installation.

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BATTERY, STORAGE, AIRCRAFT
24 VOLTS, MAINTENANCE FREE

Military Part Number:	M8565/16-1
Replaces	Not Applicable
Maximum Weight	14.6 lbs
Military “C” Rated Capacity	5.0 Ah/1.0 HR/24 °C /18 V
National Stock Number	6140-**-***-****
Contract Number	***
Lot Code	***
Date of Manufacture	***
Battery S/N	***
Manufacturer’s Name	***
Date Battery Placed in Service	***

*** Fill in applicable information

FIGURE 3. Marking for M8565/16-1 battery.

MAINTENANCE-FREE BATTERY

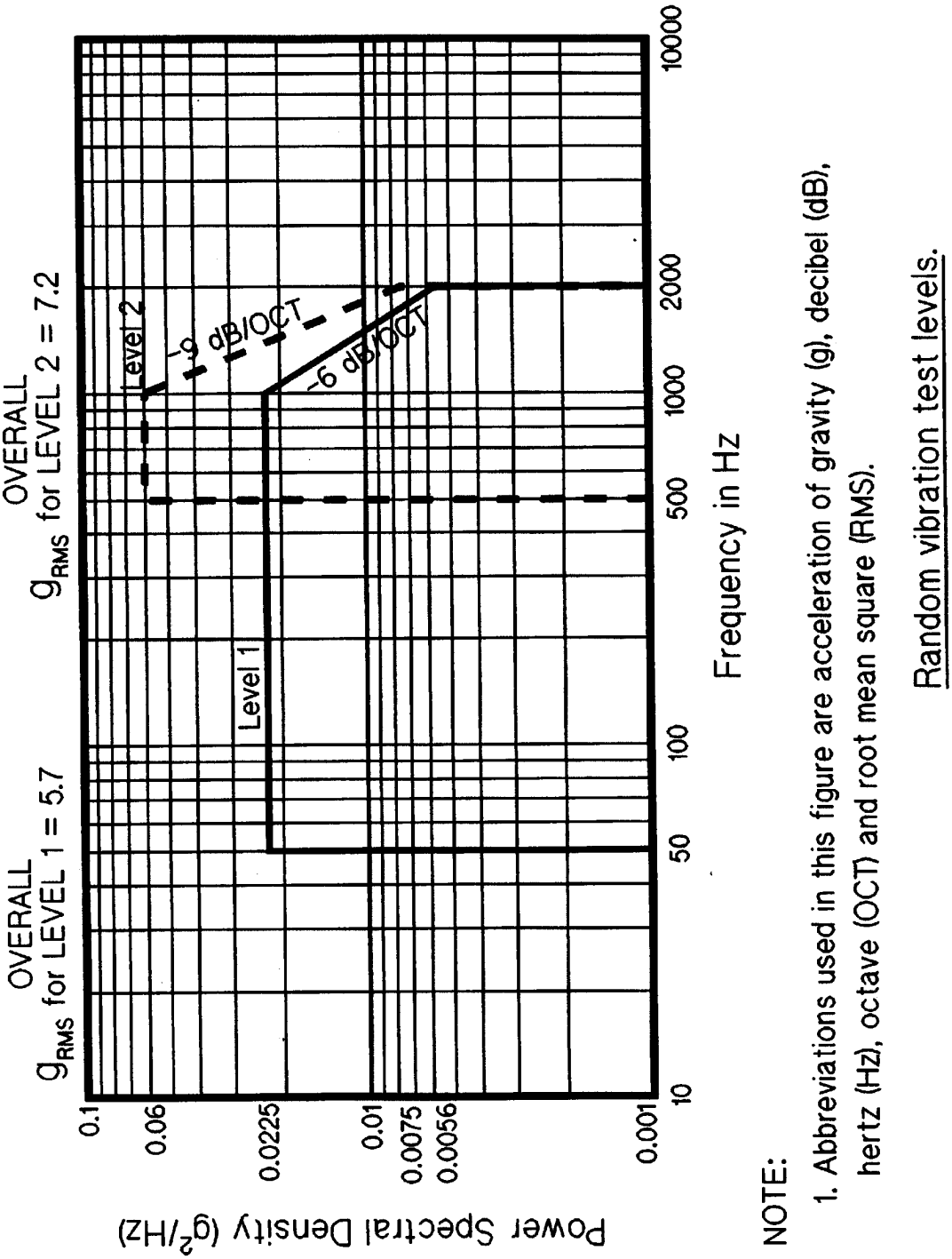
DO NOT REMOVE COVER

THIS BATTERY IS TO BE REMOVED FROM
SERVICE 36 MONTHS FROM DATE FIRST PLACED IN SERVICE

CAUTION !
TORQUE TO 35-40 IN-LB
ON HOLD-DOWN BOLTS

FIGURE 4. Battery caution marking

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Random vibration test levels.

FIGURE 5. Random vibration spectrum.

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REQUIREMENTS:

1. Part or identifying number. The part or identifying number (PIN) of the battery covered by this specification sheet shall be M8565/16-1.

2. Dimensions and weight. The dimensions of the battery shall be as shown on figure 1 contained in this specification sheet. The weight of the battery shall be not greater than 14.6 pounds (6.6 kg) (see 3.8.3 and 4.5.3).

3. Capacity. The nominal rated capacity (see 6.15.11) of the battery shall be not less than 5.0 ampere-hours (Ah).

4. Design characteristics. The battery shall use a starved-electrolyte, sealed electrochemical technology with pressure-regulated, self-resealing safety valve(s). Electrolyte leakage or spillage from the battery shall not occur during aerobatic maneuvers of the aircraft.

5. Service life. The battery's service life (useful life) in aircraft shall be not less than 36 months. The battery shall require no scheduled maintenance during the 36 months (see 3.8.9, 4.5.10, and 6.15.10).

6. Installation. Two batteries may be installed in the aircraft as shown on figure 2. The batteries may also be installed as shown in the conventional mounting example of figure 3 of MIL-PRF-8565 herein.

7. Qualification inspection and conformance inspection. Table I lists the required inspections.

8. MIL-PRF-8565 variance. The battery shall comply with MIL-PRF-8565 except as follows.

8.1 Modify the following paragraphs, tables, and figure:

3.5.4 Venting. Delete the text and substitute: “Each battery shall include two vent holes, located as shown on the applicable specification sheet. Air shall flow freely into and through the any vent holes, across the cells, and exit through the other vent holes (see 3.7.e).

3.5.5 Receptacles. Insert the following after the first sentence: “The battery's electrical interface to the aircraft shall be an MS3509 Type 1 receptacle.”

3.6.3 Identification marking. In the third sentence, delete “figure 1 herein” and substitute “figure 3 of the applicable specification sheet”.

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3.6.4 Battery caution marking. Delete the first sentence and substitute: "The battery caution marking shall contain the required titles and information as shown on figure 4 of the specification sheet and shall be placed as shown in figure 1 of the specification sheet."

3.8.5 Vent tubes. Delete.

3.8.7 Constant voltage discharge. Delete.

3.8.16 Output performance. Delete.

3.8.17 Cold and hot temperature change. Delete.

4.5.8 Constant voltage discharge. Delete.

4.5.19 Vibration test. Delete step d and insert: "Subject the battery to a resonance search over the frequency range 5 Hertz to 2,000 Hertz at 0.5*g* before and after the completion of vibration testing. No resonance shall occur within ±10 percent of 17.2 Hertz. Also subject the battery to two separate random vibration tests along each orthogonal axis. The first test shall be for a period of 20 minutes per axis to level 1 of figure 5 of the specification sheet. The second test shall be for a period of 5 minutes per axis to level 2 of figure 5 of the specification sheet". Insert the following steps:

"i. Constant potential charge the battery in accordance with 4.4.2.

"j. Discharge the battery in accordance with requirement (1) of Table III. The battery shall meet the requirement of 3.8.6 and requirement (1) of Table III."

4.5.28 Output performance. Delete.

4.5.29 Cold and hot temperature charge. Delete.

4.5.30 Wormgear retaining pin hardness. Delete.

4.5.31 Final examination. Renumber the paragraph as 4.5.36.

8.2 Add the following paragraphs:

4.5.31 Fifteen-minute discharge at 25°C. Charge the battery in accordance with 4.4.2 and stabilize it at 25 ± 2°C in accordance with 4.4.1. Discharge the battery in accordance with requirement (9) of Table III of this specification sheet, a 9C discharge for 1 minute followed immediately by a 2.5C discharge for 14 minutes. The battery shall maintain a terminal potential

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of 18.0 volts or greater throughout the total 15-minute discharge. The battery shall meet the requirement of 3.8.6 and requirement (9).

4.5.32. Three -minute discharge at -18°C. Charge the battery in accordance with 4.4.2 and stabilize it at –18 ± 2°C in accordance with 4.4.1. Discharge the battery in accordance with requirement (8) of Table III of this specification sheet. The battery shall meet the requirement of 3.8.6 and requirement (8).

4.5.33 One-minute discharge at -30°C. Charge the battery in accordance with 4.4.2 and stabilize it at –30 ± 2°C in accordance with 4.4.1. Discharge the battery in accordance with requirement (4) of Table III of this specification sheet. The battery shall meet the requirement of 3.8.6 and requirement (4).

4.5.34 Bus holdup at -20°C. Charge the battery in accordance with 4.4.2 and stabilize it at –20 ± 2°C in accordance with 4.4.1. Discharge the battery through a series resistance of 130 milliohms for a period of 55 milliseconds. During the discharge, voltage measured between the battery terminals shall be not less than 18 volts.

4.5.35 Aircraft start at -40ºC. Charge the battery in accordance with 4.4.2 and stabilize it at –40 ± 2°C in accordance with 4.4.1. Discharge the battery in accordance with requirement (6) of Table III of this specification sheet. The battery shall meet the requirement of 3.8.6 and requirement (6).

"6.11.7 Additional application information. The following information is provided for users of the M8565/16-1 battery.

"6.11.7.1 Aircraft charging system. The aircraft electrical system used for charging the battery will provide at least 50.0 ± 1.0 amperes charging current at 28.25 ± 0.50 volts (see requirement 4 of the specification sheet).

"6.11.7.2 Initial installation date. The initial installer will write the date the battery is first installed in an aircraft (date first placed in service) in the space provided on the battery's identification marking (see 3.6.3 and figure 1).

"6.11.7.3 Service life. The battery's service life (useful life) in aircraft is limited to 36 months. A battery will not be used in aircraft after 36 months from the date first installed in an aircraft (see requirement 6 of the specification sheet).

"6.11.7.4 Removal date from aircraft. The battery will be removed from aircraft service 36 months after the date it was first installed in an aircraft.

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"6.11.7.5 Uses other than for aircraft. The battery may be used for applications other than aircraft, such as ground applications, beyond the specified 36-month service life limitation if the battery is deemed satisfactory for such purposes by the user activity.

"6.11.7.6 Maintenance-free. The battery will require no periodically scheduled maintenance during its service life (see 6.15.10).

"6.11.8 Environmental characteristics. Battery performance should not be adversely affected when operating in a sand and/or dust environment. Battery materials should not support fungus growth. The battery should operate safely in an atmosphere containing ignitable or explosive fumes."

TABLE III. Capacity performance requirements. Delete the table and insert:

TABLE III. Capacity performance requirements.

Requirement number	Discharge cutoff voltage & test temperature	Discharge rate factor	Minimum time to cutoff voltage	
(1)	18.0 volts at 24 ± 2ºC	1.0C	1 hour	
(2)	18.0 volts at 49 ± 2ºC	1.0C	1.1 hour	
(3)	18.0 volts at –18 ± 2ºC	1.0C	30 minutes	
(4)	14.4 volts at –30 ± 2ºC	6.5C	1 minutes	
(5)	14.0 volts at 25 ± 2ºC	6.5C	5 minutes	
(6)	14.4 volts at –40 ± 2ºC	3.4C	2 minutes	
(7)	18.0 volts at –20 ± 2ºC	130 milliohms	55 milliseconds	
(8)	14.4 volts at –18 ± 2ºC	6.5C	3 minutes	
(9)	18.0 volts at 25 ± 2ºC	9.0C 2.5C	1 minutes 14 minutes	15 minutes total

Figure 2. Battery Caution Marking. Delete the figure.

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TABLE I. Inspection of batteries and order of test

Order of test	Description of inspection	Requirement paragraph	Test paragraph	Qualification inspection			Quality conformance inspection		
				Qualification sample			Group A	Group B tests	
				1	2	3		1	2
1	Visual & mechanical examination	3.8.2	4.5.2	X	X	X	X	X	X
2	Dimensions & weight	3.8.3	4.5.3	X	X	X	X	X	
3	Dielectric strength	3.8.4	4.5.4	X	X	X	X	X	X
4	Color & marking	3.6	4.5.2	X	X	X	X	X	X
5	Strength of vent tube	3.8.5	4.5.5	X	X	X		X	X
6	Electrical test preparation	4.4.2	4.5.6	X	X	X		X	X
7	Capacity discharge	3.8.6	4.5.7	X	X		X	X	
8	Life cycling	3.8.9	4.5.10	X				X	
9	Capacity discharge at 49°C	3.8.6	4.5.11		X				
10	Charge & discharge at low temperature	3.8.6	4.5.12			X			X
11	Discharge while inverted	3.5.9	4.5.13		X				X
12	Altitude	3.7 & 3.5.9	4.5.14		X				X
13	Mechanical shock (basic design)	3.7 & 3.8.10	4.5.15	X				X	
14	Temperature shock	3.7 & 3.8.10	4.5.16	X				X	
15	Fifteen-minute discharge	3.8.6	4.5.31		X		X		X
16	Three-minute discharge at -18ºC	3.8.6	4.5.32		X				
17	One-minute discharge at -30ºC	3.8.6	4.5.33			X			X
18	Bus holdup at -20ºC	3.8.6	4.5.34		X				X
19	Aircraft start at -40ºC	3.8.6	4.5.35		X				X
20	Temperature rise & float	3.8.11	4.5.17		X				X
21	Battery gas emission	3.8.12	4.5.18		X				
22	Vibration	3.7 & 3.8.10	4.5.19		X				X
23	Humidity	3.7 & 3.8.10	4.5.20		X				X
24	Salt fog	3.7 & 3.8.10	4.5.21	X				X	
25	Ground storage	3.8.13.1	4.5.22			X[1/]			
26	Physical integrity at high temperature	3.7 & 3.8.10	4.5.23	X					
27	Mechanical shock (crash safety)	3.8.10 & 3.8.15	4.5.25	X					
28	Impact resistance	3.8.14	4.5.24		X[2/]				
29	Deep discharge recovery	3.8.13.3	4.5.27			X			
30	Shelf life	3.8.9 & 3.8.13.2	4.5.26			X			
31	Strength of receptacle	3.8.8	4.5.9	X					
32	Final examination	3.8.4	4.5.31	X	X	X		X	X

1/ The manufacturer requesting qualification to this specification may submit an additional (fourth) test sample to expedite the qualification test program.
2/ For nonmetallic battery container designs, two outer nonmetallic battery containers must be supplied.

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Army - CR
Navy - NW
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Review activities:
Navy – AS

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Army - CR

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